Representing Theories Views about Logic

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Einführung in die nicht-klassischen Logiken (Seminar)

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Aims

- Explain the difference between inference systems and logical theories and views.
- To introduce the notion of a system of applications.
- Define the concepts of a logical theory and of a logical view based on that of a system of applications.
- Define some remarkable views and theories about logic within this framework.

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Inference systems

Definition

An *inference system* is a pair $\mathsf{S} = \langle \mathbf{Fo}_\mathsf{S}, \mathbf{Cn}_\mathsf{S} \rangle$, where $\mathbf{Fo}_\mathsf{S}$ is a set of formulae and $\mathbf{Cn}_\mathsf{S} \subseteq 2^{\mathbf{Fo}_\mathsf{S}} \times \mathbf{Fo}_\mathsf{S}$ is a consequence relation.

Examples

- classical sentential and first-order logics,
- modal logics $\mathsf{M}, \mathsf{B}, \mathsf{S4}, \mathsf{S5}$, etc.,
- Heyting's sentential intuitionistic logic, Jaskowski discussive logic, etc.

Notation

$[(\mathbf{A}, \mathtt{F}) \in \mathbf{Cn}_\mathsf{S}]$ means that $\mathtt{F} \in \mathbf{Fo}_\mathsf{S}$ is an S-logical consequence (or S-consequence) of the set of formulae in $\mathbf{A} \subseteq \mathbf{Fo}_\mathsf{S}$ – i.e., $\mathtt{F}$ follows from $\mathbf{A}$ *logically*, in the sense formalised by S.

Logical theories and views

Cf. Hjortland, 'What counts as evidence for a logical theory?', 2019.

Remark

A *theory* of logic is more than a formal inference system. A formal system, as any formalism, may be used for this or that purpose, context, topic, etc.

Desideratum

A logical theory must be an account of what system or systems count as inference systems and their purposes or contexts or topics of *correct applicability*.

Topics

Intuition

A *topic* may be understood as the content, subject matter, context (or the like) of a reasoning or an inference.

Notation

Let $\mathtt{T}, \mathtt{U}$ stand for topics and $\mathtt{S}$ for a set of topics:

$$\mathtt{T} \sqsubseteq \mathtt{U} \overset{\text{DF}}{=} \mathtt{T} \text{ is part of } \mathtt{U}$$

$$\sqcup \mathtt{S} \overset{\text{DF}}{=} \text{the fusion of all topics } \mathbb{T} \in \mathtt{S}$$

$$\mathtt{T} \sqcup \mathtt{U} \overset{\text{DF}}{=} \sqcup\{\mathtt{T}, \mathtt{U}\} \qquad \text{(the fusion of } \mathtt{T} \text{ and } \mathtt{U})$$

$$\overline{\mathcal{T}} \overset{\text{DF}}{=} \{\mathbb{T} : \mathbb{T} \text{ is a topic}\} \qquad \text{(the set of all topics)}$$

$$\overline{\mathbb{T}} \overset{\text{DF}}{=} \sqcup \overline{\mathcal{T}} \qquad \text{(the universal topic)}$$

The received view

Jan Woleński, 'The universality of logic', 2017

[T]hree understanding[s] of the universality property of logic might be discriminated:

(a) Logic is universal, if it is universally applicable;
(b) Logic is universal, if it is topic-neutral;
(c) Logic is universal, if its principles are universally true.

According to (a), every science as well as the commonsensical knowledge assume and apply logical rules, consciously or not. The view (b) denies that logic privileges any concrete subject. The point (c) claims that logical theorems are true in all circumstances, situations, states of the world, etc. (p. 23)

Statement

Various inference systems may be regarded as systems of logic, but only one (monism) – standard aka classical first-order logic (*classicalism*) – is correct, and it applies to all topics (*universalism*) in the same way (*topic-neutralism*).

Challenges to the received view

Contra classicalism:

Some views based on non-classical logics – e.g., dialetheism.

Contra monism (pluralism):

'[W]e defend what we call *logical pluralism*, the view that there is more than one genuine deductive consequence relation'. (Beall & Restall, *Logical Pluralism*, 2006, p. 3)

Contra universalism:

'In wisdom is no logic.' (Brower, 'De onbetrouwbaarheid der logische principes', 1905)

Contra topic-neutralism (topic-specificism):

There is no single logic that is independent of any content, but in each domain a logic is adequate. The logical and the physical, the formal and the real, are interdependent. (Destouches-Février, *La structure des théories physiques*, 1951, p. 88)

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Desiderata for the Notion of a Theory of logic

We want a theory of logic to state:

- which inference systems qualify as systems of logic;
- what things qualify as logical topics (i.e., those topics about which logic is topic-neutral or not);
- how those systems of logic above are correctly applicable or not to these topics.

In order to define such a notion of *theory of logic*, I will first introduce the notion of a *system of applications*.

System of applications: Definition

Definition

A *system of applications* is a triple $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \mathscr{R}_{\mathfrak{A}} \rangle$ such that:

- $\mathscr{T}_{\mathfrak{A}} = \{\mathbb{T}_1, \ldots, \mathbb{T}_l\}$ is a non-empty set of topics; and
- $\mathscr{S}_{\mathfrak{A}} = \{\mathsf{S}_1, \ldots, \mathsf{S}_m\}$ is a non-empty set of systems of logic;
- $\mathscr{R}_{\mathfrak{A}} = \{\mathbf{C}_{\mathfrak{A}}, \mathbf{R}_1, \ldots, \mathbf{R}_n\}$ is a set of relations such that $\mathbf{C}_{\mathfrak{A}}, \mathbf{R}_1, \ldots, \mathbf{R}_n \subseteq \mathscr{T}_{\mathfrak{A}} \times \mathscr{S}_{\mathfrak{A}}$, where $\mathbf{C}_{\mathfrak{A}}$ is $\mathfrak{A}$'s distinguished relation.

Pre-image and image under $\mathbf{R} \in \mathscr{R}_{\mathfrak{A}}$ and $\mathfrak{A}$

The *pre-image* $\mathbf{Pr}_{\mathbf{R}}(\mathscr{S})$ of $\mathscr{T} \subseteq \mathscr{T}_{\mathfrak{A}}$ and the *image* $\mathbf{Im}_{\mathbf{R}}(\mathscr{T})$ of $\mathscr{S} \subseteq \mathscr{S}_{\mathfrak{A}}$ under $\mathbf{R} \in \mathscr{R}_{\mathfrak{A}}$ are defined as follows:

$$\mathbf{Pr}_{\mathbf{R}}(\mathscr{S}) \overset{\text{DF}}{=} \{\mathbb{T} \in \mathscr{T}_{\mathfrak{A}} : (\mathbb{T}, \mathsf{S}) \in \mathbf{C}_{\mathfrak{A}}, \text{ for some } \mathsf{S} \in \mathscr{S}\},$$

$$\mathbf{Im}_{\mathbf{R}}(\mathscr{T}) \overset{\text{DF}}{=} \{\mathsf{S} \in \mathscr{S}_{\mathfrak{A}} : (\mathbb{T}, \mathsf{S}) \in \mathbf{C}_{\mathfrak{A}}, \text{ for some } \mathbb{T} \in \mathscr{T}\}.,$$

$$\mathbf{Pr}_{\mathfrak{A}}(\mathscr{S}) \overset{\text{DF}}{=} \mathbf{Pr}_{\mathbf{C}_{\mathfrak{A}}}(\mathscr{S}), \qquad \mathbf{Im}_{\mathfrak{A}}(\mathscr{T}) \overset{\text{DF}}{=} \mathbf{Im}_{\mathbf{C}_{\mathfrak{A}}}(\mathscr{T}).$$

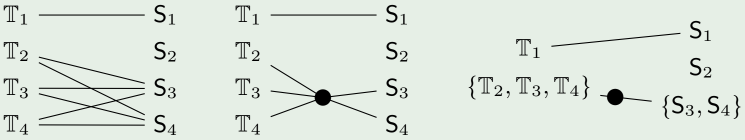
System of applications: Diagrams

Explanation

A relation $\mathbf{R} \in \mathscr{R}_{\mathfrak{A}}$ may be represented diagrammatically as a two-column graph, with the left column comprising topics, the *topics column*, and the right column comprising inference systems, *logics* or *systems column*.

Example 1

Consider the relation: $\mathbf{R} = \{\langle \mathbb{T}_1, \mathsf{S}_1\rangle, \langle \mathbb{T}_2, \mathsf{S}_3\rangle, \langle \mathbb{T}_2, \mathsf{S}_4\rangle, \langle \mathbb{T}_3, \mathsf{S}_3\rangle, \langle \mathbb{T}_3, \mathsf{S}_4\rangle, \langle \mathbb{T}_4, \mathsf{S}_3\rangle, \langle \mathbb{T}_4, \mathsf{S}_4\rangle\}$. In words, $\mathbb{T}_1$ and S_1 are $\mathbf{R}$-related exclusively, while all of $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$ are all related to all of $\mathsf{S}_3, \mathsf{S}_4$. The following diagrams are representations of $\mathbf{R}$:



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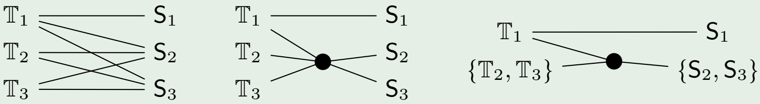
Example 2

Consider the relation: $\mathbf{R} = \{\langle \mathbb{T}_1, \mathsf{S}_1 \rangle\} \cup (\{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3\} \times \{\mathsf{S}_2, \mathsf{S}_3\})$. In words, $\mathbb{T}_1$ and S_1 are $\mathbf{R}$-related exclusively, while all of $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$ are all related to all of $\mathsf{S}_3, \mathsf{S}_4$. The following diagrams are representations of $\mathbf{R}$:



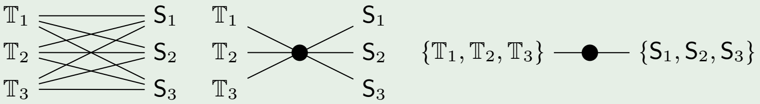
System of applications: Diagrams

Explanation

A relation $\mathbf{R} \in \mathscr{R}_{\mathfrak{A}}$ may be represented diagrammatically as a two-column graph, with the left column comprising topics, the *topics column*, and the right column comprising inference systems, *logics* or *systems column*.

Example 3

Consider the relation: $\mathbf{R} = \{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3\} \times \{\mathsf{S}_1, \mathsf{S}_2, \mathsf{S}_3\}$ In words, $\mathbb{T}_1$ and S_1 are $\mathbf{R}$-related exclusively, while all of $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$ are all related to all of $\mathsf{S}_3, \mathsf{S}_4$. The following diagrams are representations of $\mathbf{R}$:



Classes of systems of applications

Classes of systems of applications

- We can speak about a class of systems of applications by specifying some properties which are satisfied only be the members of that class.
- For instance, we might speak of all systems of applications $\mathfrak{A}$ such that $\mathscr{T}_{\mathfrak{A}}$ and $\mathscr{S}_{\mathfrak{A}}$ are singletons, and $\mathbf{C}_{\mathfrak{A}} = \mathscr{T}_{\mathfrak{A}} \times \mathscr{S}_{\mathfrak{A}}$ (the only topic is related to the only system of logic).

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Complete theory of logic: Definition

Definition

A (*complete*) *theory of logic* is the (classical) logical closure of a collection of statements stating all the following: (i) exactly which logical topics exist; (ii) exactly which mathematical systems qualify as inference systems; (iii) exactly which inference systems are correctly applicable to which logical topics.

Definition: Theory of logic based on a system of applications

A theory of logic *is based on* a system of applications $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$ iff the theory's statements is the classical logical closure of the following statements:

- $\mathbb{T}$ is a topic iff $\mathbb{T} \in \mathscr{T}_{\mathfrak{A}}$.
- S is a system of logic iff $\mathsf{S} \in \mathscr{S}_{\mathfrak{A}}$.
- S is correctly applicable to $\mathbb{T}$ iff $\langle \mathbb{T}, \mathsf{S}\rangle \in \mathbf{C}_{\mathfrak{A}}$.

Case 1: Universal classicalism

Definition

Standard classicalism is the theory (of logic) based the system of applications $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\} \rangle$, where:

- $\mathscr{T}_{\mathfrak{A}} = \overline{\mathcal{T}}$;
- $\mathscr{S}_{\mathfrak{A}} = \{\mathsf{C}\}$, where C is classical logic; and
- $\mathbf{C}_{\mathfrak{A}} = \overline{\mathcal{T}} \times \mathsf{C}$, that is,

$$\overline{\mathcal{T}} \longrightarrow \mathsf{C}$$

Example 2: Destouches-Fevrier conception

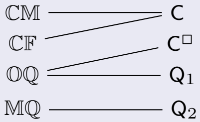
$\mathbb{CM}$: classical mechanics; $\mathbb{CF}$: classical field theory; $\mathbb{OQ}$: old quantum mechanics; $\mathbb{MQ}$: modern quantum mechanics; $\mathsf{C}^{\Box}$: an extension of C; $\mathsf{Q}_{1,2}$: two quantum logics.

[Since] a deductive theory is constructed to adequately describe a domain of science, and ... a logic must constitute the rules of reasoning of such a theory, then this logic is not independent of the content of this theory, of its domain of adequacy. ... There is no single logic that is independent of any content, but in each domain a logic is adequate. The logical and the physical, the formal and the real, are interdependent. (Destouches-Fevrier, *La structure des théories physiques*, 1951, p. 88)

Definition

D.-F.'s view is the theory (of logic) based the system of applications $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, where:

- $\mathscr{T}_{\mathfrak{A}} = \{\mathbb{CM}, \mathbb{CF}, \mathbb{OQ}, \mathbb{MQ}\}$;
- $\mathscr{S}_{\mathfrak{A}} = \{\mathsf{C}, \mathsf{C}^{\Box}, \mathsf{Q}_1, \mathsf{Q}_2\}$; and
- $\mathbf{C}_{\mathfrak{A}} = \{\langle \mathbb{CM}, \mathsf{C}\rangle, \langle \mathbb{CF}, \mathsf{C}\rangle, \langle \mathbb{OQ}, \mathsf{Q}_1\rangle, \langle \mathbb{OQ}, \mathsf{C}^{\Box}\rangle, \langle \mathbb{MQ}, \mathsf{Q}_2\rangle\}$, that is,



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Views about logic

Definition

A *view about logic* is a collection of statements stating some of the following: (i) which logical topics exist; (ii) which mathematical systems qualify as inference systems; (iii) which inference systems are correctly applicable to which logical topics.

Definition: View about logic based on a class of theories of logic

A view about logic *is based on* a class of theories of logic iff it consists of the intersection of the statements made by the theories of that class.

Remark

Views about logic cannot be represented via specific diagrams, but via schematic diagrams.

Case 1: Nihilism

Definition

Nihilism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that $\mathbf{C}_{\mathfrak{A}} = \{\}$.

Remark

Although this definition of logical nihilism appears natural, it does not reflect how this view is conceived in the literature (cf. Russell, 'Logical nihilism: could there be no logic?'). Instead, standard nihilism appears to be a variety of topic-specificism.

Diagrammatic representations

$\mathbb{T}$

S

$\mathbb{T}_1$
$\mathbb{T}_2$

S_1
S_2

$\mathbb{T}_1$
$\mathbb{T}_2$
$\mathbb{T}_3$

S

Case 2: Monism

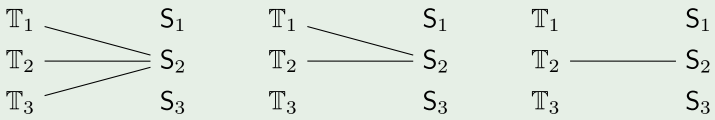
Definition

Monism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that $\mathbf{Im}_{\mathfrak{A}}$ is a singleton.

Remark

Although this definition of logical nihilism appears natural, it does not reflect how this view is conceived in the literature (cf. Russell, 'Logical nihilism: could there be no logic?'). Instead, standard nihilism appears to be a variety of topic-specificism.

Diagrammatic representations



Case 3: Pluralism

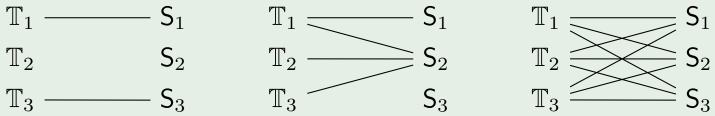
Definition

Pluralism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that $\mathbf{Im}_{\mathfrak{A}}$ has at least two members.

Remark

Note that at least two systems are connected with some topic, but there may be more, since the set $\mathscr{S}$ may be non-empty and might have some connection.

Diagrammatic representations



Case 4: Universalism

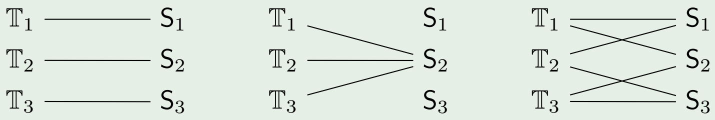
Definition

Universalism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that, for all $\mathbb{T} \in \mathscr{T}_{\mathfrak{A}}$, we have $\mathbf{Im}_{\mathfrak{A}}(\mathbb{T}) \neq \{\}$.

Remark

The notation means that all topics in $\mathscr{T}$ are connected to some logic.

Diagrammatic representations



Case 5: Anti-Universalism

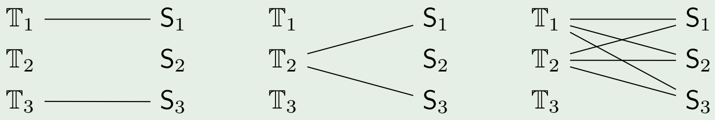
Definition

Anti-Universalism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that, there is a $\mathbb{T} \in \mathscr{T}_{\mathfrak{A}}$ with $\mathbf{Im}_{\mathfrak{A}}(\mathbb{T}) = \{\}$.

Remark

The notation means that some topic is connected to no logic.

Diagrammatic representations



Case 6: Topic-neutralism

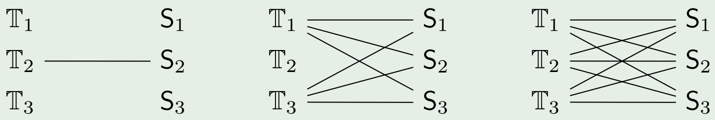
Definition

Topic-neutralism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that: (i) $\mathbf{C}_{\mathfrak{A}} \neq \{\}$ and (ii) for all $\mathbb{T}, \mathbb{U} \in \mathbf{Im}_{\mathfrak{A}}$, we have $\mathbf{Im}_{\mathfrak{A}}(\mathbb{T}) = \mathbf{Im}_{\mathfrak{A}}(\mathbb{U})$, that is.

Remark

The notation entails that topics $\mathbb{T}_1$ and $\mathbb{T}_2$ have different connections at least with respect to two systems $\mathsf{S}_1, \mathsf{S}_2$, which is enough for topic specificism.

Diagrammatic representations



Case 7: Topic-specificism

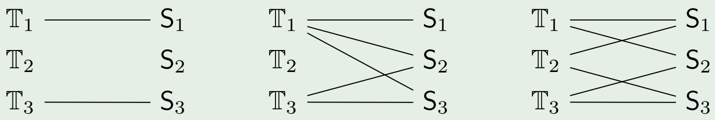
Definition

Topic-specificism is the view comprising all theories based on a system $\mathfrak{A} = \langle \mathscr{T}_{\mathfrak{A}}, \mathscr{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\}\rangle$, such that: (i) $\mathbf{C}_{\mathfrak{A}} \neq \{\}$ and (ii) there are $\mathbb{T}, \mathbb{U} \in \mathbf{Im}_{\mathfrak{A}}$ with $\mathbf{Im}_{\mathfrak{A}}(\mathbb{T}) \neq \mathbf{Im}_{\mathfrak{A}}(\mathbb{U})$.

Remark

The notation entails that topics $\mathbb{T}_1$ and $\mathbb{T}_2$ have different connections at least with respect to two systems $\mathsf{S}_1, \mathsf{S}_2$, which is enough for topic specificism.

Diagrammatic representations



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Topic-Sensitivity

Warning

Topic-specificness is not the same as topic-sensitivity (cf. Berto, *Topics of Thought*). In fact, the latter is compatible with topic-neutrality.

Topic-sensitivity

Topic-sensitivity is a property of some inference systems which make them distinguish whether two formulae may be about two different topics. Consider, e.g., the following sentences:

1. It is raining or it is not raining.
2. Laura is a musician or Laura is not a musician.

Given their distinct topics, however, these sentence are not logically equivalent in a topic-sensitive system.

In sum, *topic-sensitivity is a property of inference systems and not of logical views, like topic-neutrality.*

Conclusions

1. An inference system, by itself, is not a theory of logic.
2. A complete theory of logic determines:
   - the domain of logical topics;
   - the class of acceptable inference systems;
   - the applicability relation between topics and systems.
3. The notion of a system of applications provides a framework for representing the relation between topics and systems of logic within a view about logic.
4. The framework distinguishes between (fully stated) theories of logic and families of views about logic.

$$Ευχαριστώ!$$

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